COMPUTER SYSTEMS AND PLATFORM TECHNOLOGIES

IT0047

FEU ALABANG FEU DILIMAN FEU TECH

Technology Driven by Innovation

MODULE 3

DOS & DEBUG Commands

MODULE 3.1

DOS Commands

At the end of the lesson, the student should be able to :

- **Define MSDOS**
- **Identify the types of DOS Commands**
- **Identify the most used DOS Commands**

Learning MS-DOS Basics - A Tutorial

This tutorial gives you an opportunity to try basic MS-DOS commands. By following the procedures in this section, you will learn to:

- View the contents of a directory
- Change from one directory to another
- Create and delete directories
- Change from one drive to another
- Copy files o Rename files
- Delete files o Format a floppy disk

The Command Prompt

When you first turn on your computer, you will see some cryptic information flash by. MS-DOS displays this information to let you know how it is configuring your computer. You can ignore it for now. When the information stops scrolling past, you'll see the following:

C:\>

This is called the command prompt or DOS prompt. The flashing underscore next to the command prompt is called the cursor. The cursor shows where the command you type will appear.

Typing a Command

This section explains how to type a command at the command prompt and demonstrates the "Bad command or file name" message.

1. To type a command at the command prompt

Type the following at the command prompt (you can type the command in either uppercase or lowercase letters):

nul

If you make a typing mistake, press the BACKSPACE key to erase the mistake, and then try again.

Typing a Command

2. Press ENTER

You must press ENTER after every command you type.

The following message appears:

Bad command or file name

The "Bad command or file name" message appears when you type something that MSDOS does not recognize. Because nul is not a valid MS-DOS command, MS-DOS displays the "Bad command or file name" message.

Typing a Command

3. Now, type the following command at the command prompt:

ver

The following message appears on your screen:

MS-DOS version 6.22

Viewing the Contents of a Director

In this section, you will view the contents of a directory by using the dir command. The dir command stands for "directory.“

- To view the contents of a directory

1. Type the following at the command prompt:

dir

A list similar to the following appears:

```
Volume in drive C is MS-DOS_6.22
Volume Serial Number is 1E49-15E2
Directory of C:\

WINDOWS      <DIR>    09-08-92 10:27p
TEMP         <DIR>    05-15-92 12:09p
CONFIG  SYS       278 09-23-92 10:50a
COMMAND COM     53014 09-18-92 6:00a
WINA20  386      9349 11-11-91 5:00a
DOS          <DIR>    09-02-92 4:23p
AUTOEXEC BAT      290 09-23-92 10:54a

              7 file(s)  62931 bytes
                       8732672 bytes free
```

Viewing the Contents of a Director

This is called a directory list. A directory list is a list of all the files and subdirectories that a directory contains. In this case, you see all the files and directories in the main or root directory of your drive. All the files and directories on your drive are stored in the root directory.

Changing Directories

Look at the list on your screen. All the names that have beside them are directories. You can see a list of the files in another directory by changing to that directory, and then using the dir command again. In this case, you will change to the DOS directory

Before you begin this section, make sure you have a directory named DOS by carrying out the following procedure

- To make sure you have a directory named Windows

  If you do not see a line in the directory list indicating that you have a directory named Windows, type the following at the command prompt:

  dir /s Windows

- You will see a message that includes a line such as the following:

  Directory of C:\DIRNAME

Viewing the Contents of a Director

To change from the root directory to the WINDOWS directory

- To change directories, you will use the cd command. The cd command stands for "change directory.“
- Type the following at the command prompt:

cd windows

- The command prompt changes. It should now look like the following:

C:\WINDO

Next, you will use the dir command to view a list of the files in the DOS directoryWS>

Viewing the Contents of a Director

To view a list of the files in the WINDOWS directory

1. Type the following at the command prompt:

Dir

A list of the files in the WINDOWS directory appears, but scrolls by too quickly to read. You can modify the dir command so that it displays only one screen of information at a time.

To view the contents of a directory one screen at a time

Viewing the Contents of a Director

- Type the following at the command prompt:

dir /p

One screen of information appears. At the bottom of the screen, you will see the following message:

Press any key to continue

- To view the next screen of information, press any key on your keyboard. Repeat this step until the command prompt appears at the bottom of your screen

Viewing the Contents of a Director

When you typed the dir command this time, you included the /p switch after the command. A switch modifies the way MS-DOS carries out a command. Generally, a switch consists of a forward slash (/) that is followed by one or more letters or numbers. When you used the /p switch with the dir command, you specified that MS-DOS should pause after it displays each screen of directory list information. The p actually stands for "page

Another helpful switch you can use with the dir command is the /w switch. The /w switch indicates that MS-DOS should show a wide version of the directory list.

Viewing the Contents of a Director

o To view the contents of a directory in wide format

1. Type the following at the command prompt:

Dir

The directory list appears, with the filenames listed in wide format. Note that only filenames are listed. No information about the files' size or date and time of creation appears.

/w

If the directory contains more files than will fit on one screen, you can combine the /p and /w switches as follow

dir /w /p

Changing Back to the Root Directory

Next, you will change from the DOS directory to the root directory. The root directory is the directory you were in before you changed to the DOS directory. Before you begin this section, make sure your command prompt looks like the following:

C:\DOS>

o To change to the root directory

1. Type the following at the command prompt:

cd \

Viewing the Contents of a Directory

Note that the slash you type in this command is a backslash (\), not a forward slash (/)

No matter which directory you are in, this command always returns you to the root directory of a drive. The root directory does not have a name. It is simply referred to by a backslash (\).

The command prompt should now look like the following:

C:\>

When your command prompt appears similar to this---that is, when it does not contain the name of a directory---you are in the root directory.

Creating a Directory

In this section, you will create two directories. Creating a directory is helpful if you want to organize related files into groups to make them easy to find. Before you begin this section, make sure the command prompt looks like the following

C:\>

To create a directory, you will use the md command. The md command stands for "make directory."

To create and change to a directory named FRUIT

Creating a Directory

Type the following at the command prompt:

md fruit

You have now creat ed a directory named FRUIT. You won't see the new FRUIT directory until you carry out the dir command in the next step.

2. To confirm that you successfully created the FRUIT directory, type the following at the command prompt:

Dir

Or

dir /p

Creating a Directory

To create and work with a directory named GRAPES

Type the following at the command prompt:

md grapes

You will not see the new GRAPES directory until you carry out the dir command in the next step.

To confirm that you successfully created the GRAPES directory, type the following at the command prompt:

Creating a Directory

```
dir
```

A list similar to the following appears:

```
Volume in drive C is MS-DOS-6
Volume Serial Number is 1E49-15E2
Directory of C:\FRUIT

.           <DIR>        09-25-93 12:08p
..          <DIR>        09-25-93 12:08p
GRAPES      <DIR>        09-25-93 12:10p
                   3 file(s) 0 bytes
                     11534336 bytes free
```

Note that there are three entries in the FRUIT directory. One is the GRAPES directory that you just created. There are two other entries---one looks like a single period (.) and the other looks like a double period (..). These directory entries are important to MSDOS, but you can ignore them. They appear in every directory and cont ain information

Creating a Directory

The GRAPES directory is a subdirectory of the FRUIT directory. A subdirectory is a directory within another directory. Subdirectories are useful if you want to further subdivide information.

To change to the GRAPES directory, type the following at the command prompt:

cd grapes

The command prompt should now look like the following:

C:\FRUIT\GRAPES>

Creating a Directory

. To switch back to the FRUIT directory, type the following:
cd ..
The command prompt should now look like the following
C:\FRUIT>
When the cd command is followed by two periods (..), MS-DOS moves up one level in the directory structure. In this case, you moved up one level from the GRAPES directory to the FRUIT directory

Deleting a Directory

If you no longer use a particular directory, you may want to delete it to simplify your directory structure. Deleting a directory is also useful if you type the wrong name when you are creating a directory and you want to delete the incorrect directory before creating a new one.

In this section, you will delete the GRAPES directory. Before you begin this section, make sure the command prompt looks like the following:

C:\FRUIT>

To delete a directory, use the rd command. The rd command stands for "remove directory."

To delete the GRAPES directory

Deleting a Directory

To delete the GRAPES directory
Type the following at the command prompt:
rd grapes
To confirm that you successfully deleted the GRAPES directory, type the following at the command prompt:
dir
The GRAPES directory should no longer appear in the directory list.
Note You cannot delete a directory if you are in it. Before you can delete a directory, you must make the directory that is one level higher the current directory. To do this, type cd.. at the command prompt.

Changing Drives

This section describes how to change drives. Changing drives is useful if you want to work with files that are on a different drive.
So far, you have been working with driveC. You have other drives you can use to store information. For example, drive A is your first floppy disk drive. The files and directories on drive A are located on the floppy disk in the drive. (You might also have a drive B, which contains the files and directories stored on the floppy disk in that drive.)
Before you begin this section, make sure your command prompt looks like the following:
C:\FRUIT>

Changing Drives

To change to and view files on a different drive

1. Insert a 3.5" floppy disk in drive A label- side up. Make sure the disk clicks into the drive.

Type the following at the command prompt:

a:

Note that the command prompt changed to the following:

A:\>

This message may appear:

Not ready reading drive

A Abort, Retry, Fail?

Changing Drives

If you see this message, the disk may not be inserted properly. Place the disk label-side up in the disk drive, and make sure the disk clicks into the disk drive. Then, type r for Retry. If this message appears again, press F for Fail, and then type b: at the command prompt. If you no longer see this message, type b: instead of a: throughout the rest of the tutorial.

There must be a floppy disk in the drive that you want to change to.

Changing Drives

Change back to drive C by typing the following at the command prompt:

c:

Your command prompt should return to the following:

C:\FRUIT>

When you type a drive letter followed by a colon, you change to that drive. The drive letter that appears in the command prompt shows which drive is the current drive. Unless you specify otherwise, any commands you type are carried out on the current drive and in the current directory

Copying Files

This section describes how to copy a single file and a group of files. Copying files creates a duplicate of the original file and does not remove the original file. This is useful for many reasons. For example, if you want to work on a document at home, you can copy it from your computer at work to a floppy disk and then take the floppy disk home. To copy a file, you will use the copy command. When you use the copy command, you must include two parameters. The first is the location and name of the file you want to copy, or the source. The second is the location to which you want to copy the file, or the destination. You separate the source and destination with a space. The copy command follows this pattern:

Copying Files

copy source destination

Copying a Single File

In this section, you will copy the notepad.exe files from the WINDOWS directory to the FRUIT directory. You will specify the source and destination of these files in two different ways. The difference between the two methods is explained at the end of this section.

Before you begin this section, make sure the command prompt looks like the following:

C:\FRUIT>

To copy the NOTEPAD.EXE files from the WINDOWS directory to the FRUIT directory

Copying Files

Return to the root directory by typing the following at the command prompt:

cd\

The command prompt should now look like the following:

C:\>

r

Change to the DOS directory by typing the following at the command prompt:

cd windows

Make sure the file you are going to copy, NOTEPAD.EXE, is located in the WINDOWS directory by using the dir command followed by a filename.

dir notepad.exe

A list similar to the following appears:

Copying Files

To copy the NOTEPAD.EXE file from the WINDOWS directory to the FRUIT directory, type the following at the command prompt:
copy c:\windows\notepad.exe c:\fruit
The following message appears:
1 file(s) copied r

To confirm that you copied the files successfully, view the contents of the FRUIT directory by typing the following at the command prompt:
dir \fruit
You should see the file listed in the FRUIT directory

Renaming Files

This section explains how to rename files. You may want to rename a file if the information in it changes or if you decide you prefer another name.

To rename a file, you will use the ren command. The ren command stands for "rename." When you use the ren command, you must include two parameters.

The first is the file you want to rename, and the second is the new name for the file. You separate the two names with a space. The ren command follows this pattern:

ren oldname newname

- Renaming a File

In this section, you will rename the README.TXT file

Renaming Files

Before you begin this section, make sure your command prompt looks like the following:

C:\FRUIT>

1. To rename the NOTEPAD.EXE file to PADNOTE.TXT, type the following at the command prompt:

ren notpad.exe padnote.txt

2. To confirm that you renamed the file successfully, type the following at the command prompt:

dir

Deleting Files

This section explains how to delete, or remove, a file that you no longer want on your disk. If you don’t have very much disk space, deleting files you no longer use is essential.

To delete a file, you will use the del command. The del command stands for "delete." r

• Deleting a File

In this section, you will delete two files using the del command. Before you begin, make sure your command prompt looks like the following:

C:\FRUIT>

Deleting Files

To delete the PEARCOM and PEAR.HLP files

1. Delete the PADNOTE.TXT file by typing the following at the command prompt:

del PADNOTE.TXT

2. To confirm that you deleted the files successfully, type the following at the command prompt:

dir

• Deleting a Group of Files

In this section, you will use wildcards to delete a group of files.

Before you begin this section, make sure your command prompt looks like the following:

C:\FRUIT>

Deleting Files

To delete files in the current directory that end with the extension OLD by using wildcards

1. View all files that end with the extension OLD by typing the following at the command prompt:

dir *.old r

A list of all the files that end with the extension OLD appears. Make sure that these are the files you want to delete. When you are deleting files by using wildcards, this step is very important. It will prevent you from deleting files accidentally.

2. Delete all files ending with OLD by typing the following at the command prompt: del *.old 3. To confirm that all the files with the extension OLD have been deleted, type the following at the command prompt:

Deleting Files

2. Delete all files ending with OLD by typing the following at the command prompt:
del *.old
3. To confirm that all the files with the extension OLD have been deleted, type the following at the command prompt:
dir
The FRUIT directory should contain no files.
Now that the FRUIT directory is empty, you can delete it by using the rd (remove directory) command that you learned to use in "Deleting a Directory" earlier in this chapter.

Deleting Files

2. Delete all files ending with OLD by typing the following at the command prompt:
del *.old
3. To confirm that all the files with the extension OLD have been deleted, type the following at the command prompt:
dir
The FRUIT directory should contain no files.
Now that the FRUIT directory is empty, you can delete it by using the rd (remove directory) command that you learned to use in "Deleting a Directory" earlier in this chapter.

Changing Drives

To delete the FRUIT directory

1. Return to the root directory by typing the following at the command prompt:

cd \

2. You can see the FRUIT directory in the directory list by typing the following at the command prompt:

dir or dir /p

3. Remove the FRUIT directory by typing the following at the command prompt:

rd fruit

Changing Drives

To verify that the FRUIT directory has been removed, type the following at the command prompt:
dir or dir /p
The FRUIT directory should not appear in the directory list.

MODULE 3.2

DEBUG

At the end of the lesson, the student should be able to :

- **Define DEBUG**
- **Identify the most used DEBUG Commands**
- **Write a program with string manipulation using DEBUG**

Debug

DOS offers a debugging utility as one of its external commands. The DEBUG program is used for testing and debugging executable programs. A feature of DEBUG is that it displays all program code and data in hexadecimal format, and any data that you enter into memory must also be in hex format. DEBUG also provides a single-step mode, which allows you to execute a program one instruction at a time, so that you can view the effect of each instruction on memory locations and registers.

Rules of DEBUG command

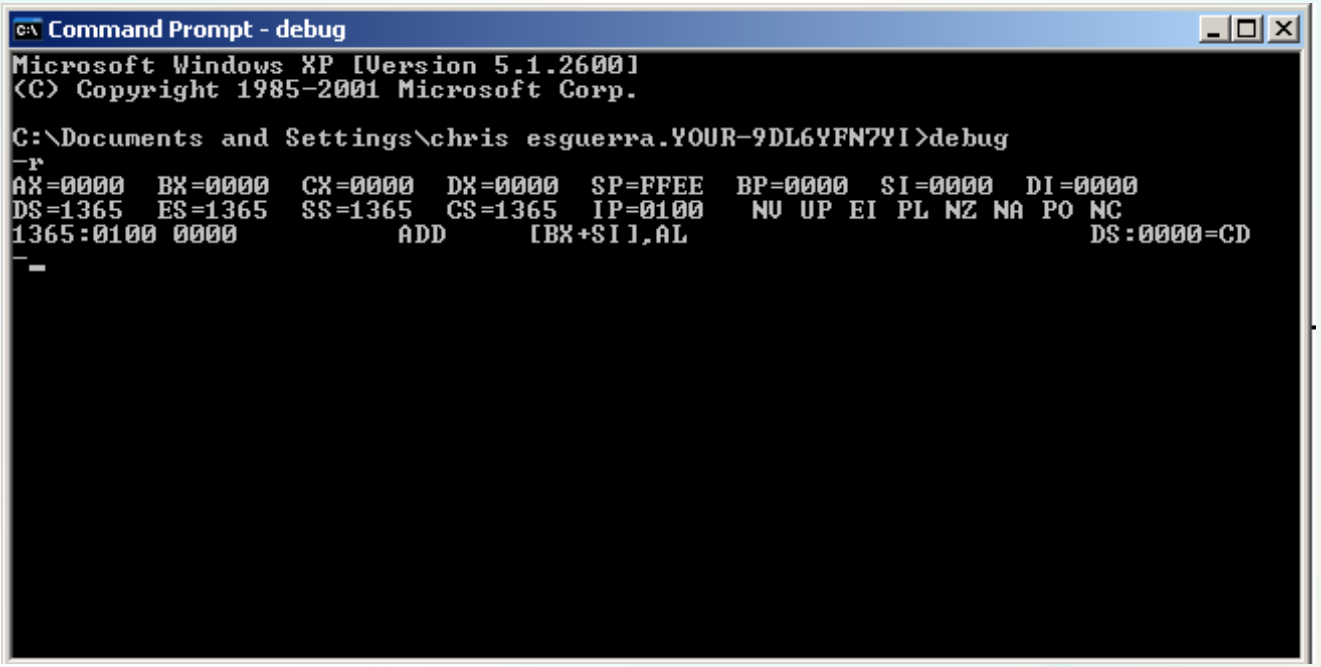
- The prompt from the DEBUG program is a hyphen (-).
- A command is a single letter, usually followed by one or more parameters. Commands become effective only after ENTE00000000000R key is pressed.
- Commands are not case sensitive, can be uppercase or lowercase letter or combination of both.
- Delimiters may separate commands and parameters. Delimiters are only required, however, between two consecutive hexadecimal values.

DEBUG commands

r

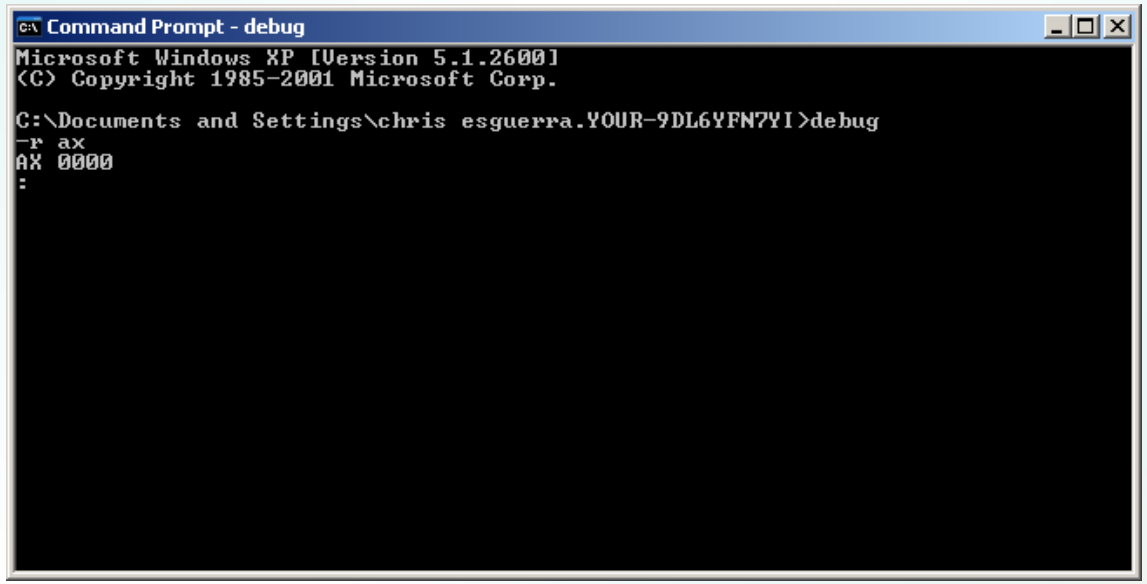
Internal Registers

- **R** - Display the hexadecimal contents of all the register, and the flag register current status and the next instruction to be executed.



Internal Registers

- R <registername> - Display the contents of internal register and with the option of changing those contents.

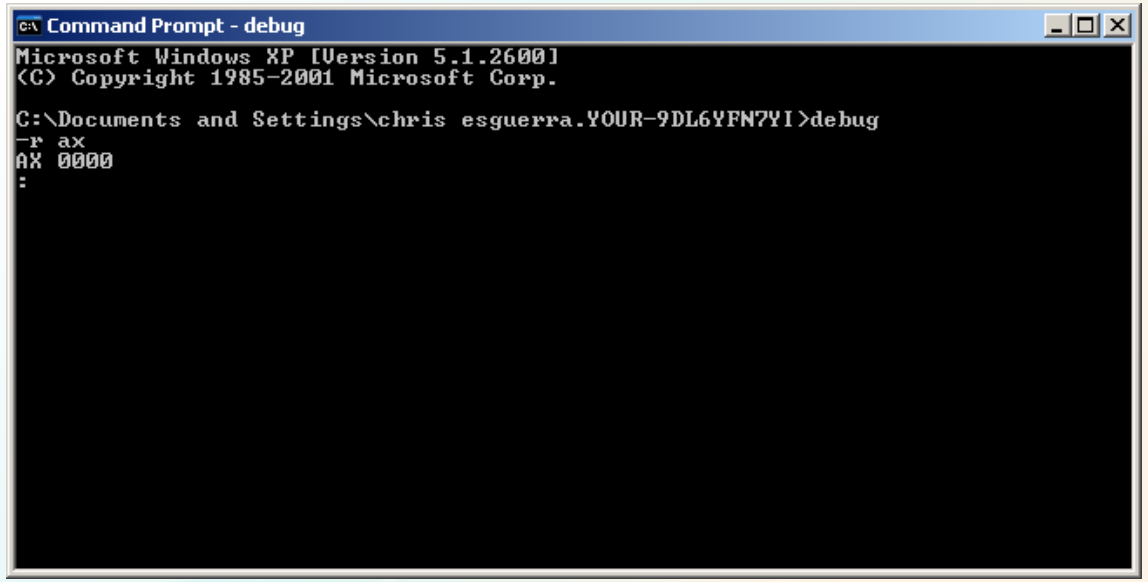


Machine Language Instruction

- The last line of the Register(R) command indicates the location in memory of the next instruction to be executed and its machine mnemonic code (unassembled) codesThe E (Enter) Command displays and allows modification of bytes in a sequential manner.

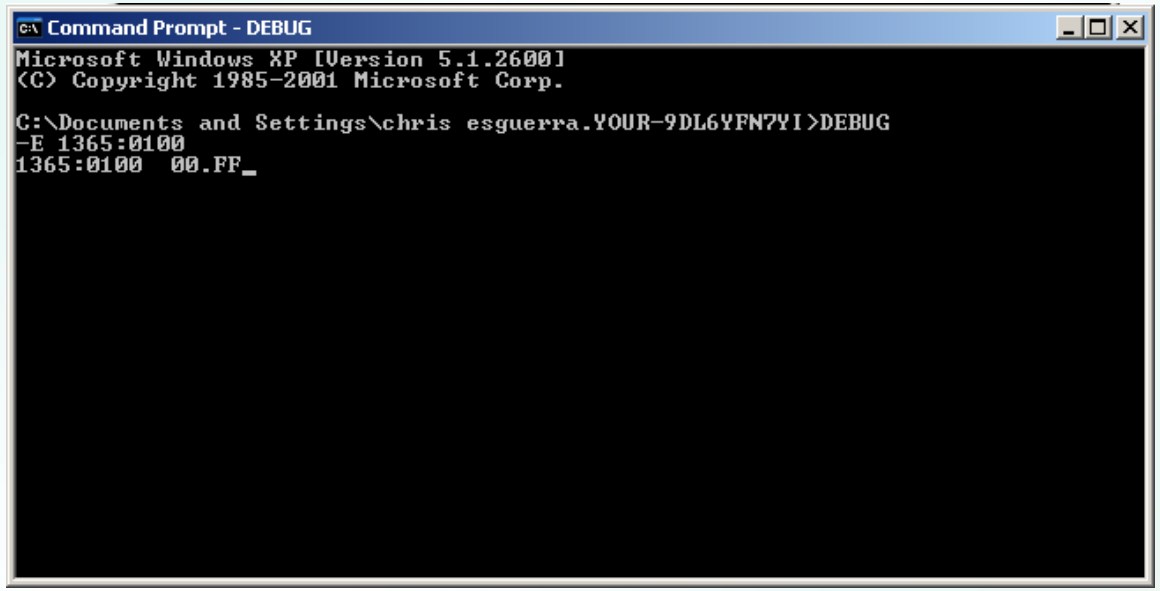
Format: E <address>

Note: Address should contain the code segment and instruction pointer where you want to modify.



Machine Language Instruction

To change the content of the address specified above, from 19h to FFh, enter FF after the dot (.).



Mnemonic Codes

- The U (unassembled) command translates the contents of the memory into its equivalent mnemonic code assembler – like statements and displays their addresses.

  Format: U <address>

```
-u 1365:0100
1365:0100 01C0          ADD     AX,AX
1365:0102 01D8          ADD     AX,BX
1365:0104 01C8          ADD     AX,CX
1365:0106 01D0          ADD     AX,DX
1365:0108 29C0          SUB     AX,AX
```

Unassembled from the first address to last address specified in Enter Command activity and translate each machine code into its equivalent mnemonic code. Write down on the data sheet the results of this activity.

Program Execution

Writing Down Program in DEBUG Environment

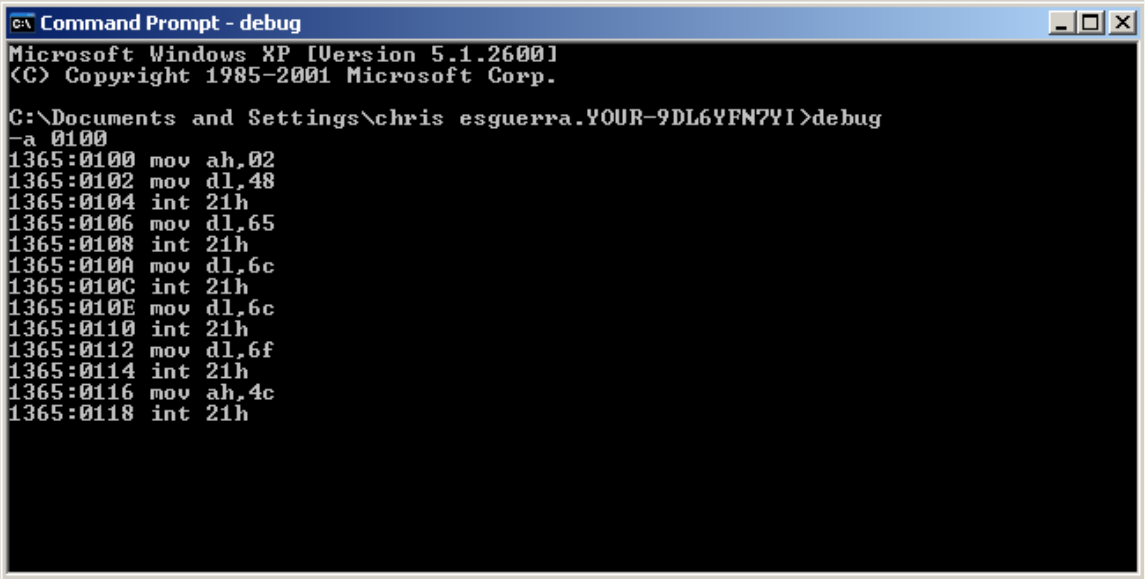
The A instructions.

(Assemble) command lets you enter mnemonic code assembly language

Format: A <address>

- Assume for the address that will serve as the offset address.

Assemble the following code into memory location 1365:0100.



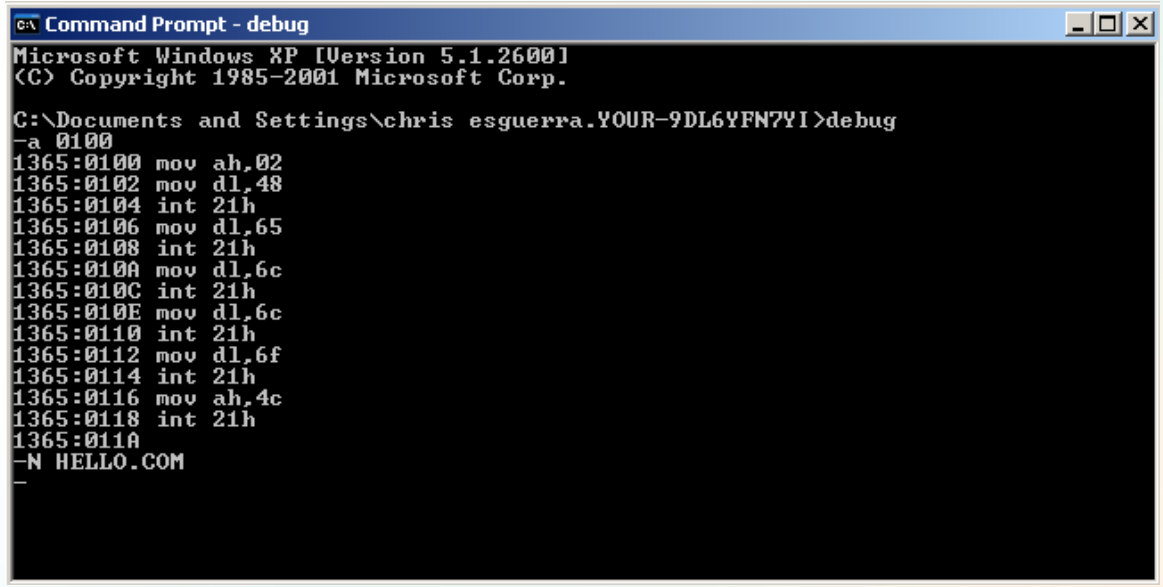
Program Execution

Saving the Program into the Disk

To write a file, you must first initialize the name of the file.

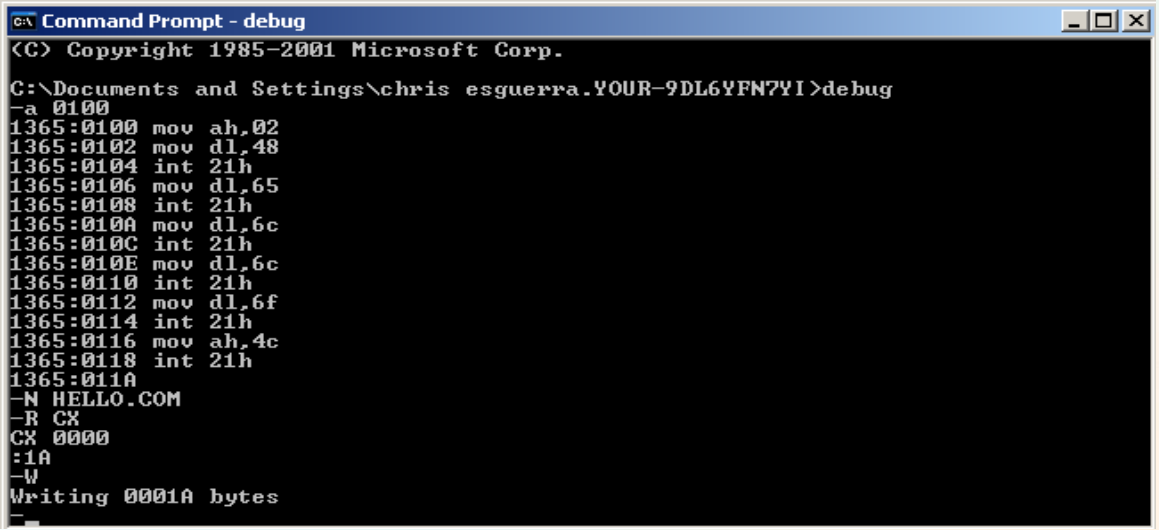
Format: N <filename>.<ext>

The N command initializes a filename in memory before using load or write command.



Program Execution

In writing the program to the disk, computer for the byte consumed by the program then, store it in CX (Count) Register. To compute, subtract the starting address to the last address covered by the program plus 2.

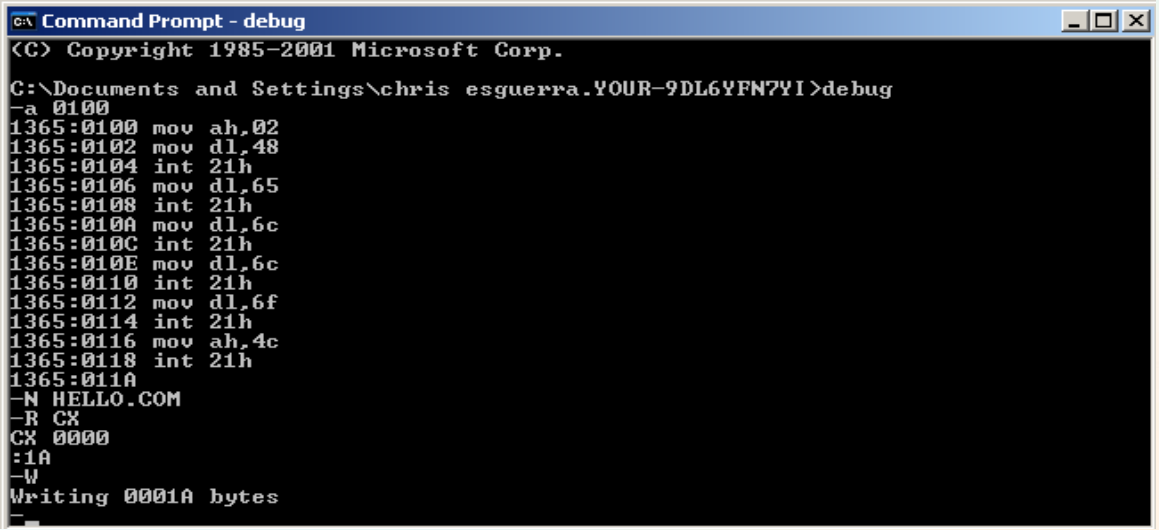


Note: Computation is in hexadecimal Arithmetic Operation

Consumed byte = (0118 – 0100) + 2

Program Execution

In writing the program to the disk, computer for the byte consumed by the program then, store it in CX (Count) Register. To compute, subtract the starting address to the last address covered by the program plus 2.



Note: Computation is in hexadecimal Arithmetic Operation

Consumed byte = (0118 – 0100) + 2

The W (write) command writes a block of memory to a file.

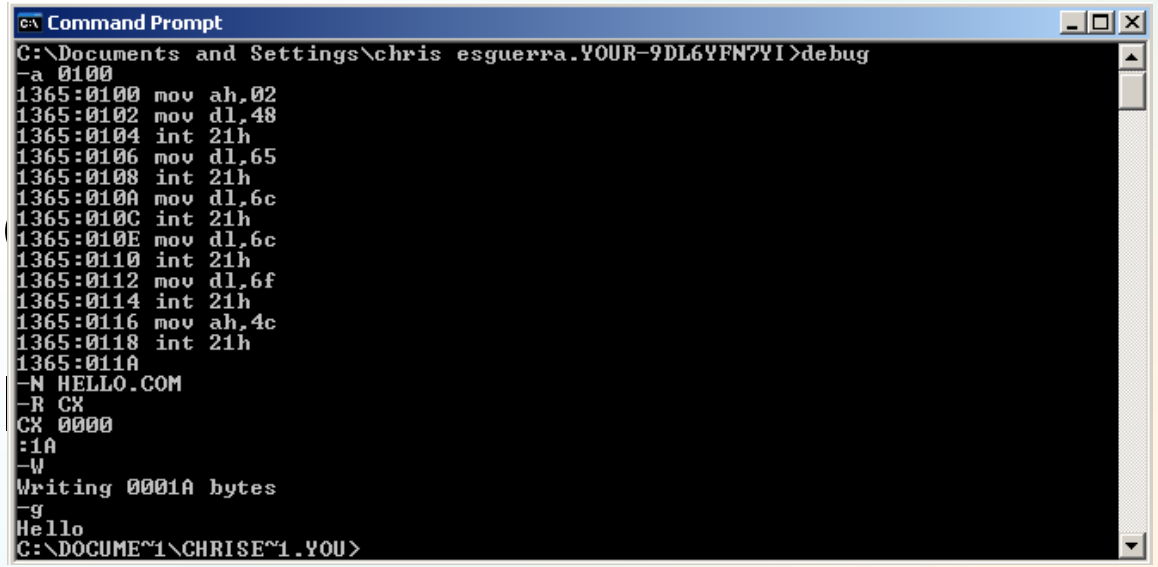
Program Execution

Executing the Program

Using the G (GO) command, it will execute the assembled program.

Format: G

The T (trace) command executes one more instructions starting with the instruction at CS:IP. The contents of all register and flags after each execution are also displayed.



Format: T

DOS FUNCTION CALLS

To make the task of programming in Assembly Programming easier, DOS provides a set of instruction for character device I/O, file management, memory management, data and time functions, execution of the other programs, and others. This set of instructions is assigned as interrupt vector 21h, and each function can be called upon by the INT 21H instruction.

DOS FUNCTION CALLS

Using DOS Function

DOS Function INT 21H (“Function Dispatcher”) requires the function number in register AH on entry only. Most of the function calls require input to be passed to them in the other register.

1. Set the proper register values required.
2. Place the function number in AH.
3. Issue interrupt type 21H.

TRADITIONAL CHARACTER DEVICE INPUT AND OUTPUT

Function : Direct Console Input without Echo
Purpose : Waits for a character to be read from the keyboard then returns with the character in AL.
On Entry : AH = 07H
On Exit : Character read from the keyboard.
Function : Character Output
Purpose : Outputs a character to the standard output device, usually monitor screen. On Entry : AH = 02H
DL = ASCII code of the character to be output
On Exit : NONE

TRADITIONAL CHARACTER DEVICE INPUT AND OUTPUT

Function : Character Input with Echo
Purpose : Waits for a character to be read from the keyboard, then echoes it to the screen and return with the character in AL.
On Entry : AH = 01H
On Exit : AL = Character read from the keyboard

String Manipulation (DEBUG)

Displaying strings on the screen and reading them from the keyboard is entirely different from displaying or reading characters. ASCII codes of all characters are only a byte, which would easily fit inside any of the internal registers of the 8088. A function that displays a character only needs the character to be displayed to be put in the DL register, e.a. function 02h. Similarly, a function that reads a character from the keyboard can easily get the character read in the register AL, e.a. functions 01h and 07h.

String Manipulation (DEBUG)

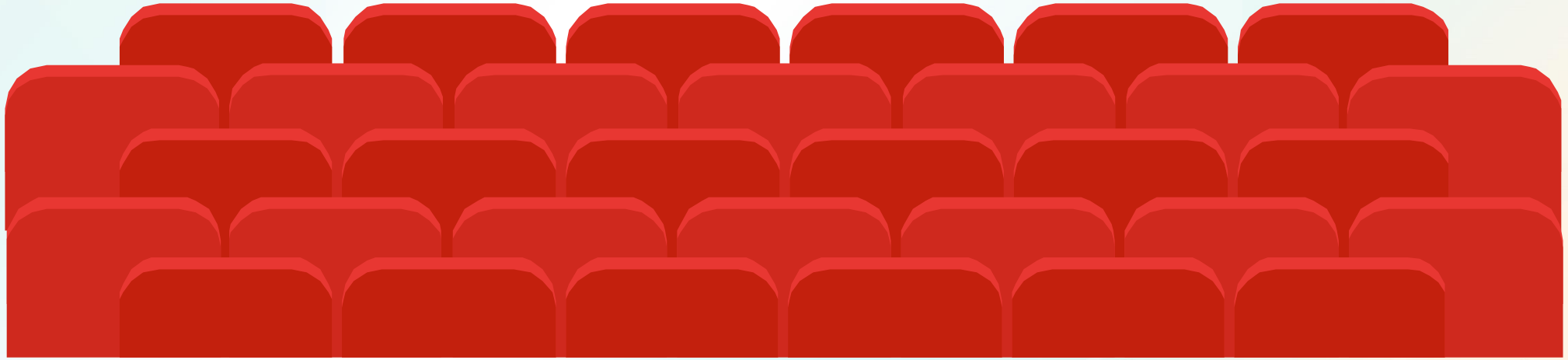
A string is usually more than one character length. This leaves us with the question: “Where then can a function that displays a string put the string to be displayed?” The answer is
– the memory. Since string’s can’t usually fit in a register, strings are then placed in the memory then pass the address of the string in memory to two of the registers – the segment address and offset address in DX.

String Manipulation (DEBUG)

Function : String Output
Purpose : Sends a string of characters to the standard output device On Entry : AH = 09h
DS = Segment address of the first character of the string DX = Offset address of the first character of the string
On Exit : NONE

Note: Function 09h displays a string of characters starting with the first character (address in DS:DX) output, but not including the character “$” (ASCII -24h)

Q&A SESSION

ASK ANY QUESTION RELATED TO OUR TOPIC FOR TODAY.

THANK YOU